Person: researcher
runs the pipeline

External system: Mendeley Data V1
raw and cleaned workbooks

HTTPS

Software system: Task 3 analysis repository

Container: fetch_data.py
download + SHA-256 check

M1: sha256 match

Data store: data/raw/
workbooks (ignored),
test_run.jsonl

Container:
process_data.prepare
coding, eligibility, UPS
scoring

M3: latency_ns, phase,
status

M2: exclusions, 300/300
totals

Data store: data/processed/
coded CSVs,
metrics_summary.csv

M5: p50/p95/p99, e, d,
speed-up

Container: collector.py
warm-up + benchmark runs

Container:
process_data.run_analysis
statistics and aggregation

M4: rho, CI, p, R2,
guardrails

Data store: reports/
results.json, figures

M6: dpi = 300, PDF present

Container: visualize.py
figures PDF + 300 DPI PNG

Container:
01_data_analysis.ipynb
executed report